EVRYN Evidence Grounding Contract: Dual-Span Provenance & Causal Directionality

Cause Entity (C)

Canonical: PmrB

Mention: "PmrB sensor kinase"

Entity Type: Sensor Kinase / Protein

Section: Results | Page: 2

RELATION: DIRECT_CAUSE (+)

Direction: FORWARD | Confidence: 0.95 | Polarity: POSITIVE

Effect Entity (E)

Canonical: PmrA

Mention: "response regulator PmrA"

Entity Type: Response Regulator / Protein

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Primary Evidence Sentence & Verbatim Provenance Coordinates (S_causal)

100% PROVENANCE VERIFIED

"Acid pH activates the PmrA/PmrB system by promoting PmrB-mediated phosphorylation of PmrA."

Doc ID: PMC1804205 | Section: RESULTS | Paragraph: 14 | Sentence ID: s42 | Offsets: [1420–1512] | Page: 2

Intra-Sentence Context

1. Direct Intra-Sentence Assertion — Single Sentence

- Both C & E co-occur in S_causal boundary
- Syntactic trigger: "activation of", "induces"
- Coordinates: <p.2, Results, [1420–1512]>
- High empirical precision (0.90–0.98 on gold)

Anaphoric Context Resolution

2. Context-Resolved Candidate — Cross-Sentence

- Inter-sentence alignment across [S_k, S_k+1]
- Demonstrative pronoun & anaphora resolution
- Dual provenance binding: [S_cause, S_effect]
- Abbreviation expansion via domain normalizer

Transitive Multi-Hop Linking

3. Multi-Page Transitive Link — Multi-Page Graph

- Path traversal: A -> B (p.2) -> C (p.8)
- Inter-page distance: Δp >= 1, hops <= 2
- Pruned by domain invariants (TCSTS_REVERSE_PAIRS)
- Distance decay: conf = prod(c_i) * λ^d